FLAT PULLBACK COMMUTES WITH THE Q FUNCTOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Parasitic kernel and cokernel of the adjunction map. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-adjoint-kernel-parasitic (2). Identifies the comparison map's defect.
- Flat pullback preserves parasitic modules. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-parasitic (1). Keeps the defect parasitic after pullback.
- Parasitic cohomology exactness criterion. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-exact-sequence-quasi-coherent-parasitic-cohomology. Parasitic kernel and cokernel in QCoh imply the map is an isomorphism.

Source reference. Original lemma 0778: sources/stacks-chow/tags.txt, source lines 9193–9193. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06WH: sources/stacks-chow/tags.txt, source lines 8816–8816. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original proposition 0771: sources/stacks-chow/tags.txt, source lines 9186–9186. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0779: sources/stacks-chow/tags.txt, source lines 9194–9194. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0774: sources/stacks-chow/tags.txt, source lines 9189–9189. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0776: sources/stacks-chow/tags.txt, source lines 9191–9191. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a flat morphism of algebraic stacks. Then $Q_{\mathcal{X}} \circ f^* = f^* \circ Q_{\mathcal{Y}}$ where $Q_{\mathcal{X}}$ and $Q_{\mathcal{Y}}$ are as in Lemma 0778.*

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Proof. Observe that f^* preserves both $QCoh$ and $LQCoh^{fbc}$, see Sheaves on Stacks, Lemma 06WH and Proposition 0771. If $\mathcal{F}$ is in $LQCoh^{fbc}(\mathcal{O}_\mathcal{Y})$ then $Q_\mathcal{Y}(\mathcal{F}) \to \mathcal{F}$ has parasitic kernel and cokernel by Lemma 0779. As f is flat we get that $f^*Q_\mathcal{Y}(\mathcal{F}) \to f^*\mathcal{F}$ has parasitic kernel and cokernel by Lemma 0774. Thus the induced map $f^*Q_\mathcal{Y}(\mathcal{F}) \to Q_\mathcal{X}(f^*\mathcal{F})$ has parasitic kernel and cokernel and hence is an isomorphism for example by Lemma 0776. $\square$